Test Ninjas GMAT

GMAT Math Formulas Cheat Sheet for the Focus Edition

This GMAT math formulas cheat sheet is built for the Focus Edition: every formula tested in the 21-question, 45-minute Quant section, organized so you can drill it in one sitting. No geometry, no calculator — just arithmetic and algebra, the exact way GMAC now tests them. Bookmark it, print it, or work through the interactive formula lookup below.

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GMAT Formula Lookup

Pick a topic to surface the exact formula and a one-line reminder of when to use it.

CONCEPT FORMULA

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What the GMAT Focus Edition Actually Tests

Before memorizing a single identity, know what you are memorizing *for*. The GMAT Focus Edition rewrote the Quantitative Reasoning section in 2023, and most of the "cheat sheets" circulating online still teach the old test. If your GMAT math formulas cheat sheet still leads with geometry, it is already out of date.

Section structure at a glance

The Focus Edition Quant section gives you 21 multiple-choice problem-solving questions to answer in 45 minutes. That is roughly 2 minutes 8 seconds per question, which is why automatic formula recall matters so much — every second you spend reconstructing a formula is a second you cannot spend thinking about the trap.

Core structural facts for the 2026 GMAT Focus Edition Quantitative Reasoning section.

Attribute	Detail
Question count	21 problem-solving questions
Time limit	45 minutes (about 2:08 per question)
Topics tested	Arithmetic and algebra only
Removed from Quant	Geometry, data sufficiency
Calculator	Not permitted (scratch paper only)

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Section score scale	60–90
Top possible section score	90 (≈100th percentile)

What got removed (and what got harder)

Geometry disappeared from Quant entirely — no more circle areas, cone volumes, or coordinate plane distance formulas to memorize. Data sufficiency questions also moved to the new Data Insights section. What remains is a denser focus on arithmetic and algebra, where you cannot hide behind a diagram. Expect more layered percent-change problems, more weighted averages, more number-property puzzles, and more algebraic manipulation than the legacy test ever asked.

The no-calculator rule

The on-screen calculator appears only in the Data Insights section. In Quant you get scratch paper and whatever mental math you walked in with. That constraint is the whole reason a formula cheat sheet matters — you cannot brute-force a compound-interest question; you have to know the formula and plug numbers efficiently.

> **Key takeaway:** Half the formulas on older GMAT cheat sheets are irrelevant to the Focus Edition. Throw out the geometry — drill the arithmetic and algebra cold.

Arithmetic Formulas: The High-Volume Section

Arithmetic is where most Focus Edition Quant questions live. These GMAT quant formulas cover percentages, interest, averages, and ratios — concepts that feel elementary but get twisted into multi-step word problems. Start here.

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Percent change	(new − old) / old × 100	Price hikes, salary raises, population growth
Percent of a number	part = percent × whole / 100	Discounts, commission, markup
Simple interest	I = Prt	Loan or deposit over a fixed rate
Compound interest	A = P(1 + r/n)^(nt)	Savings accounts, repeated compounding
Arithmetic mean	sum / count	Averages of test scores, sales, speeds (when equal weights)
Weighted average	Σ(value × weight) / Σ(weight)	Mixed groups, GPA-style problems
Ratio to fraction	a : b = a / (a + b) of the whole	Part-to-whole ratio conversions

Percent change and percent of

Percent change is the single highest-frequency formula on the test. The standard form is (new − old) / old × 100, and the trap is always the denominator — if the question asks for the percent change from A to B, the denominator is A, not B. Successive percent changes multiply: a 20% increase followed by a 10% decrease is 1.20 × 0.90 = 1.08, or a net 8% increase.

Simple and compound interest

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number of compounding periods per year. If the problem says "compounded annually," n = 1. "Quarterly" means n = 4, and "monthly" means n = 12 — unit conversion is where careless mistakes happen.

Averages and weighted averages

The arithmetic mean is just sum over count. The weighted average, however, respects group size: Σ(value × weight) / Σ(weight). If Class A averages 80 with 30 students and Class B averages 90 with 10 students, the combined average is (80 × 30 + 90 × 10) / 40 = 82.5, *not* 85. Mixing up weighted with simple averages is one of the classic GMAT traps.

Ratios and proportions

A ratio a : b represents parts of a whole. If a class has boys and girls in a 3 : 5 ratio and 40 students total, then boys = 3/(3+5) × 40 = 15. Cross-multiplication solves any proportion a/b = c/d by a × d = b × c — the fastest route through most ratio word problems.

WORKED EXAMPLE — SUCCESSIVE PERCENT CHANGE

Setup: A jacket is marked up 25% above cost, then discounted 20% at the register. If the final register price is $120, what did the jacket cost the store?

1. Let cost = C. After a 25% markup, listed price = 1.25C.
2. After a 20% discount, final price = 0.80 × 1.25C = 1.00C.
3. So 1.00C = $120, meaning C = $120.
4. Sanity check: 25% up then 20% down is a net 0% — the classic percent-change trap.

> **Result:** The jacket cost the store $120 — the same as the register price, because a 25% increase followed by a 20% decrease nets to zero.

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The single highest-frequency arithmetic formula on the GMAT. Enter old and new values to see the percent change.

OLD VALUE

e.g. 100

NEW VALUE

e.g. 125

Calculate

QUESTION 1 — PERCENT CHANGE

A stock price rises 20% in January and then falls 10% in February. What is the net percentage change over the two months?

- +8%
- +10%
- +12%
- +2%
- −2%

Check Answer

Algebra Formulas: Equations, Exponents, and Inequalities

Algebra is the other half of the Focus Edition Quant section. These GMAT algebra formulas cover linear and quadratic equations, exponent and root rules, absolute value, and slope. Every

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Concept	Formula	Common Mistake
Quadratic formula	$x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$	Dropping a sign under the radical
Difference of squares	$a^2 - b^2 = (a + b)(a - b)$	Forgetting it applies to any two squared terms
Perfect square trinomial	$a^2 \pm 2ab + b^2 = (a \pm b)^2$	Missing the middle term's sign
Exponent product	$a^m \cdot a^n = a^{m+n}$	Adding bases instead of exponents
Power of a power	$(a^m)^n = a^{mn}$	Adding instead of multiplying exponents
Negative exponent	$a^{-n} = 1 / a^n$	Treating the negative like subtraction
Slope	$m = (y_2 - y_1) / (x_2 - x_1)$	Inverting rise and run
Absolute value	$\|x\| = x$ if $x \geq 0$, else $-x$	Forgetting the two cases when solving $\|x + a\| = b$

The quadratic formula and factoring

The quadratic formula $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$ solves any $ax^2 + bx + c = 0$. Before invoking it, always try to factor first — if you spot the difference of squares ($a^2 - b^2$) or a perfect square trinomial, factoring is seconds faster than the full formula. When factoring fails, check the discriminant ($b^2 - 4ac$) for sign: negative means no real roots, zero means one repeated root, positive means two distinct roots.

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exponents), $(a^m)^n = a^{mn}$ (power of a power → multiply), and $a^m / a^n = a^{m-n}$ (divide → subtract). Negative exponents flip the term: $a^{-n} = 1/a^n$. Fractional exponents map to roots: a^(1/n) = $\sqrt[n]{a}$. The GMAT loves combining these in one question to see if you can chain them without losing a sign.

Absolute value and inequalities

Absolute value is just distance from zero on the number line. Solving $|x + a| = b$ always splits into two cases: $x + a = b$ or $x + a = -b$. With inequalities, flip the direction when you multiply or divide both sides by a negative — that single rule traps more students than anything else in this section.

Slope and linear equations

Slope $m = (y_2 - y_1) / (x_2 - x_1)$. Point-slope form $y - y_1 = m(x - x_1)$ and slope-intercept form $y = mx + b$ cover every linear-equation question on the test. Parallel lines share the same slope; perpendicular lines have slopes that multiply to −1.

WORKED EXAMPLE — QUADRATIC FORMULA

Setup: Solve $2x^2 - 5x - 3 = 0$ using the quadratic formula.

1. Identify $a = 2$, $b = -5$, $c = -3$.
2. Compute the discriminant: $b^2 - 4ac = 25 - 4(2)(-3) = 25 + 24 = 49$.
3. Apply the formula: $x = [5 \pm \sqrt{49}] / (2 \cdot 2) = [5 \pm 7] / 4$.
4. Split the ± into two roots: $x = 12/4 = 3$ or $x = -2/4 = -1/2$.

> **Result:** $x = 3$ or $x = -1/2$. Always double-check the sign on c — sign errors under the radical are the single most common quadratic-formula mistake.

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What are the roots of $3x^2 - 7x + 2 = 0$?

- x = 2 or x = 1/3
- x = −2 or x = −1/3
- x = 3 or x = 1/2
- x = 7 or x = 2
- x = 2 only

Check Answer

Number Properties and Divisibility Rules

Number properties are less "formulas" and more "instant filters" — they let you rule out answer choices without computing. Every test-taker should be able to eyeball whether a number is divisible by 2, 3, 4, 5, 6, 9, or 10 in under a second.

Fast mental-math filters for common divisors — no scratch work required.

Divisor	Rule	Example
2	Last digit is even (0, 2, 4, 6, 8)	346 → ends in 6 → divisible
3	Digit sum is divisible by 3	147 → 1+4+7 = 12 → divisible
4	Last two digits are divisible by 4	5,312 → 12 ÷ 4 = 3 → divisible
5	Last digit is 0 or 5	2,075 → ends in 5 → divisible

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6	Divisible by both 2 and 3	222 → even and 2+2+2=6 → divisible
9	Digit sum is divisible by 9	729 → 7+2+9 = 18 → divisible
10	Last digit is 0	4,130 → ends in 0 → divisible

Divisibility tests you can eyeball

The rules above collapse long division into a glance. They are worth drilling until they are reflexive, because many GMAT questions hinge on recognizing that a number is (or is not) divisible by a small integer. A single divisibility check can eliminate two or three answer choices instantly.

Primes, LCM, and GCD

A prime number has exactly two positive divisors: 1 and itself. The primes below 20 (2, 3, 5, 7, 11, 13, 17, 19) should be memorized. Two identities tie the least common multiple and greatest common divisor together: LCM(a, b) × GCD(a, b) = a × b. Prime factorization is the engine behind both — break each number into its prime factors, take the maximum power of each prime for the LCM and the minimum for the GCD.

Odd/even and sign rules

Odd + odd = even. Even + even = even. Odd + even = odd. Odd × odd = odd. Any product with an even factor is even. Sign rules mirror this: negative × negative = positive, negative × positive = negative. These rules can answer "must be odd/even" or "must be positive/negative" questions in seconds without computing actual values.

> **Remember:** Number-property rules are not really formulas — they're instant filters. Drill them until they trigger on sight so you stop wasting time re-deriving them.

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GMAT statistics formulas cover mean, median, mode, standard deviation (conceptually), and range. Probability and counting cover permutations, combinations, and the addition and multiplication rules. These are the compact formulas most students remember least well — drilling them pays off disproportionately.

Statistics, counting, and probability formulas every Focus Edition Quant test-taker should know.

Concept	Formula	When to Use
Mean	sum / count	Equally weighted average of a set
Median	Middle value of an ordered set	When outliers would distort the mean
Range	max − min	Simple spread measure
Permutations	$P(n, r) = n! / (n - r)!$	Order matters (e.g., finishing positions)
Combinations	$C(n, r) = n! / [r!(n - r)!]$	Order does not matter (e.g., team picks)
Probability	P(A) = favorable / total	Single event probability
P(A or B)	P(A) + P(B) − P(A and B)	Events that can overlap
P(A and B), independent	P(A) × P(B)	Two independent events both happening

Mean, median, mode, and range

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most often. The range is the difference between the largest and smallest values. Watch for problems that ask which measure of central tendency an outlier would change — the mean always moves, the median usually does not.

Standard deviation intuition

The GMAT rarely asks you to compute a standard deviation by hand. What you do need is the intuition: standard deviation measures how spread out the data is around the mean. A set clustered tightly around the mean has a low SD; a set with values far from the mean has a high SD. If every value increases by a constant, the SD does not change. If every value is multiplied by k, the SD is multiplied by |k|.

Permutations vs combinations

The single question to ask: does order matter? If yes, use permutations P(n, r) = n! / (n − r)! — think seating arrangements, race finishes, lock codes. If no, use combinations C(n, r) = n! / [r!(n − r)!] — think committees, teams, lottery picks. Mixing them up is the single biggest counting mistake on the test.

Probability addition and multiplication

For mutually exclusive events (cannot both happen), P(A or B) = P(A) + P(B). For overlapping events, subtract the overlap: P(A or B) = P(A) + P(B) − P(A and B). For independent events (one does not affect the other), P(A and B) = P(A) × P(B). The complement rule — P(not A) = 1 − P(A) — is often the fastest path through an "at least one" probability question.

WORKED EXAMPLE — ALL-WOMEN COMMITTEE

Setup: A committee of 3 is chosen at random from 5 men and 4 women. What is the probability that the committee is all women?

1. Total ways to pick 3 from 9 people: C(9,3) = 9! / (3! · 6!) = 84.
2. Ways to pick 3 women from 4: C(4,3) = 4! / (3! · 1!) = 4.
3. Probability = favorable / total = 4 / 84.

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> **Result:** P(all women) = 1/21, or about 4.8%. Order does not matter here, so combinations — not permutations — are the right tool.

QUESTION 4 — COMBINATIONS

How many different 3-person committees can be formed from a group of 7 people?

- 21
- 35
- 42
- 84
- 210

Check Answer

Word Problem Templates: Rate, Work, and Mixture

Most Focus Edition Quant questions are word problems in disguise. Locking in a template for each common setup — rate, work, mixture, average speed — saves precious seconds on test day.

Distance = Rate × Time

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boat in a current), sum or difference the rates depending on direction, then apply d = rt once.

Combined work rate

When two workers (or pipes, or machines) work together, add their rates: 1/t = 1/a + 1/b. If Alice finishes a job in a hours alone and Bob in b hours alone, the combined time t satisfies that equation. The most common error is averaging the two times (3 hours and 6 hours = 4.5 hours together — wrong). You always add *rates*, never average times.

Mixture and average-speed traps

Mixture problems are weighted averages. If you mix 4 liters of 20% acid solution with 6 liters of 50% acid solution, the blended concentration is (4 × 20 + 6 × 50) / (4 + 6) = 38%. Average speed is the same trap in motion form: average speed = total distance / total time, not the arithmetic mean of two speeds. If a car drives 60 mph for half the distance and 40 mph for the other half, the average speed is 48 mph, not 50.

WORKED EXAMPLE — COMBINED WORK RATE

Setup: Pipe A fills a tank in 6 hours. Pipe B fills the same tank in 4 hours. If both pipes run together, how long does it take to fill the tank?

1. Pipe A's rate = 1/6 tank per hour.
2. Pipe B's rate = 1/4 tank per hour.
3. Combined rate = 1/6 + 1/4 = 2/12 + 3/12 = 5/12 tank per hour.
4. Time = 1 / (5/12) = 12/5 hours = 2 hours 24 minutes.

> **Result:** Together the pipes fill the tank in 2 hours 24 minutes. The combined-rate formula 1/t = 1/a + 1/b is the fastest setup for any 'two workers together' problem.

QUESTION 3 — COMBINED WORK

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- 1 hour
- 1.5 hours
- 2 hours
- 2.5 hours
- 4.5 hours

Check Answer

How to Actually Memorize (and Use) These Formulas

A GMAT equation guide is only as useful as your ability to recall it under pressure. Passive re-reading is the worst study method; active recall paired with worked problems is the best. Here is what actually works.

Derive before you drill

Before memorizing a formula, derive it once from first principles. Why does the compound interest formula multiply (1 + r/n) by itself nt times? Because each compounding period adds r/n of the principal, and nt is the number of periods. Once you understand the mechanics, the formula is nearly impossible to forget — and if you do blank on test day, you can rebuild it.

Hand-written flashcards beat apps

Digital flashcard apps are convenient, but the evidence favors hand-written cards for durable memory. The act of writing forces you to encode the formula, and retrieving it from a physical

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Pair every formula with one problem

A formula in isolation is useless; a formula attached to a worked example is portable. For each card you write, solve one GMAT-style problem that uses the formula. That single link between recall and application is what transforms memorized formulas into test-day tools.

Keep a running error log

Every time you miss a practice question, log it: the question, the correct answer, the formula involved, and the specific step you got wrong. After two weeks you'll see patterns — maybe you consistently flip percent-change direction, or drop signs in the quadratic formula. That log is the highest-leverage study document you can produce.

Formula Mastery Checklist

0/7 complete

- [ ] **Hand-wrote flashcards for all 40+ core formulas**
  Writing the card yourself strengthens recall.
- [ ] **Derived each formula at least once from first principles**
  If you can derive it, you can rebuild it on test day.
- [ ] **Paired every formula with one worked problem**
- [ ] **Drilled a mixed problem set (not siloed by topic)**
- [ ] **Set up a running error log tagged by formula**
- [ ] **Timed myself on 10 consecutive problems at 2:08 each**
  Matches Focus Edition Quant pacing.
- [ ] **Reviewed all divisibility rules this week**

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Optional deep-dives

What about Pythagorean triples and special triangles?

How do I factor quickly when I forget the quadratic formula?

Are there any formulas specific to Data Insights?

Frequently Asked Questions

Do I need to memorize geometry formulas for the GMAT Focus Edition?

No. The GMAT Focus Edition removed geometry from the Quantitative section entirely. The 21 problem-solving questions now test only arithmetic and algebra. You still need some geometric reasoning if a Data Insights question references a chart or table, but dedicated geometry formulas like circle area or cone volume are no longer required.

Can I use a calculator on GMAT Quant?

No. The on-screen calculator is available only in the Data Insights section. In the 45-minute, 21-question Quantitative Reasoning section you must compute everything mentally or on scratch paper. That is why mental-math speed and clean formula recall matter so much — every extra second on arithmetic is a second you cannot spend thinking.

How many formulas do I actually need to memorize?

Most high scorers use roughly 40 to 60 core formulas, heavily concentrated in arithmetic, algebra, number properties, and statistics. A comprehensive cheat sheet looks long, but many entries (sign rules, exponent rules, divisibility tests) become automatic once you drill them. Focus on understanding each formula well enough to derive it rather than memorizing it cold.

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On the GMAT Focus Edition, each section, including Quantitative Reasoning, is scored on a 60–90 scale. A perfect section score of 90 places you at roughly the 100th percentile. The legacy GMAT used a 6–51 scale where 51 was the top Quant score. Official percentile bands are updated annually by GMAC.

What is the fastest way to memorize GMAT math formulas?

Hand-written flashcards combined with spaced repetition beat passive re-reading. Group formulas by topic, quiz yourself daily, and pair each formula with one worked problem so recall triggers application. Test-prep authorities consistently point to active retrieval — writing the cards yourself and solving problems with them — as the most durable method.

Are formula shortcuts worth using on the GMAT?

Yes, when they are truly shortcuts. Knowing Pythagorean triples (3-4-5, 5-12-13), special-triangle ratios, and the difference-of-squares pattern can shave 30 to 60 seconds off a problem. But shortcuts you half-remember cost more time than they save — only use a shortcut you have drilled until it feels automatic.

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Study Plan
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Essay Coach
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